

Risques climatiques — M2 Actuariat

# Climate model: emissions, concentrations and temperature

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## Objectives

- Understanding the **basics** of climate models to predict world temperatures.
- Simulating climate models on historical data and make some forecasts conditional on emission scenarios.

## Materials:

- `Climate_Models_Notebook.ipynb` *notebook introduction of economic models.*
- `Climate_Models.py` *package to simulate reduced form climate models.*

# Introduction

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# Intro: Why should economists care about climate science?

- Climate change is driven by physical processes that can be **summarized in a simple causal chain**.
- Integrated Assessment Models (IAMs) like DICE or FAIR rely on this chain.
- To interpret or extend IAMs, we need to understand the building blocks.

## The Climate-Economy Chain

Emissions → Concentrations → Radiative Forcing → Temperature → Impacts

- Each arrow corresponds to a physical process.
- IAMs implement reduced-form versions of these processes.

# Intro: Climate science is about big models

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How do climate scientists study the system?

- **General Circulation Models (GCMs)** and **Earth System Models (ESMs)**:
  - Simulate atmosphere, ocean, cryosphere, land surface, and biogeochemical cycles.
  - Built on physical laws (fluid dynamics, thermodynamics, radiative transfer).
  - Used in IPCC reports for future climate projections.
- Strengths:
  - High fidelity, spatial detail, many processes included.
  - Provide benchmarks for scientific understanding.
- Limitations for Integrated Assessment Modelling:
  - Extremely computationally intensive (supercomputers).
  - Results not easily integrated into optimization or policy models.

## Today's focus: Economics of climate change

- We will work with **Reduced-form Climate Models (RCMs)**:
  - Tractable and transparent → usable in economic models.
  - Capture the key physical mechanisms without full Earth System complexity.

## The Climate-Economy Chain

Emissions → Concentrations → Radiative Forcing → Temperature

**Roadmap:** We will follow this chain step by step, from *emissions inputs* to *temperature outcomes*, highlighting at each stage how reduced-form models approximate the underlying processes.

## Emissions (the inputs)

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## Emissions: What are emissions?

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- **Definition:** Anthropogenic greenhouse-gas (GHG) *flows* into the atmosphere (mainly CO<sub>2</sub>, CH<sub>4</sub>, N<sub>2</sub>O).
- **Focus today:** CO<sub>2</sub> (fossil fuel + land use). Others can be added (e.g., CH<sub>4</sub>) as separate inputs.
- Emissions are flows  $\Rightarrow$  they feed atmospheric *stocks* in the carbon cycle.



# Emissions: Three main green house gases

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The three main sources:

- **Carbon dioxide ( $\text{CO}_2$ )**
  - From fossil fuel combustion, cement, land-use change.
  - Long-lived: persists for centuries.
  - $\sim 76\%$  of global GHG emissions ( $\text{CO}_2\text{-eq}$ ).
- **Methane ( $\text{CH}_4$ )**
  - From agriculture (rice, livestock), fossil fuel extraction, waste.
  - Short lifetime ( $\sim 10$  years) but high warming power.
  - $\sim 16\%$  of global GHG emissions.
- **Nitrous oxide ( $\text{N}_2\text{O}$ )**
  - From fertilizers, biomass burning, industry.
  - Lifetime  $\sim 120$  years.
  - $\sim 6\text{--}7\%$  of global GHG emissions.

To summarize this section, you only need to remember:

- In principle, models can take three distinct exogenous inputs:

$$\{E_t^{CO_2}\}_{t \geq 0}, \quad \{E_t^{CH_4}\}_{t \geq 0}, \quad \{E_t^{N_2O}\}_{t \geq 0}.$$

- In simplified reduced-form models, only CO<sub>2</sub> emissions are modeled explicitly; CH<sub>4</sub> and N<sub>2</sub>O are often represented by an additional exogenous forcing term.

# Concentrations

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## Concentrations: from to emissions to concetration

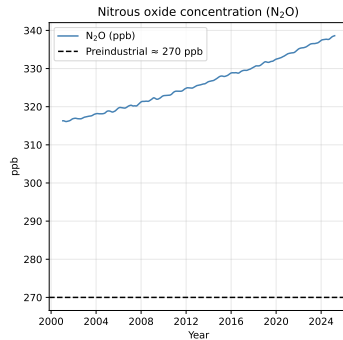
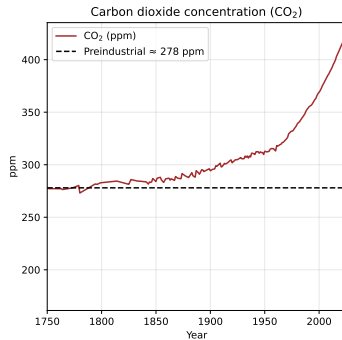
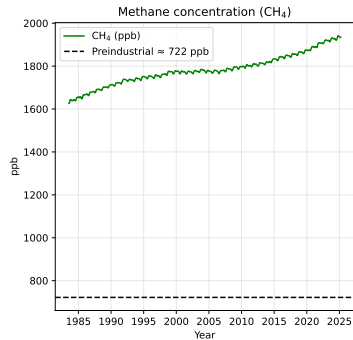
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[Emissions (flows)]  $\rightarrow$  [Atmospheric Concentration (stock)]  $\rightarrow \dots$

From emissions (flows) to concentrations (stocks):

- Climate does not react directly to annual emissions, but to the **cumulative stock of gases** in the atmosphere.
- Emissions are measured in flows (e.g. GtC/yr).
- Once emitted, gases accumulate in the atmosphere.
- What matters for climate forcing is the concentration (ppm or ppb).

# Cnocentrations: Atmospheric concentrations



# Concentrations: The mathematics of concentrations

## Matrix difference equation (step $\Delta$ ):

$$\mathbf{M}_{g,t+\Delta} = \mathbf{M}_{g,t-1} + \Delta \mathbf{A}_g (\mathbf{M}_{g,t-1} - \mathbf{M}_{g,0}) + \Delta \mathbf{b}_g E_t^g$$

- $\mathbf{M}_{g,t} \in \mathbb{R}^{n_g}$ : vector of compartment stocks/concentrations at  $t$ .
- $\mathbf{M}_{g,0}$ : baseline (natural equilibrium) vector (e.g. preindustrial levels).
- $\mathbf{A}_g \in \mathbb{R}^{n_g \times n_g}$ : *transfer matrix* (diagonal entries  $< 0$ , off-diagonals  $\geq 0$ ); row sums  $\leq 0$  reflect net losses to sinks.
- $\mathbf{b}_g \in \mathbb{R}^{n_g}$ : injection vector indicating which box receives anthropogenic emissions.
- $E_t^g$ : emissions flow (units depend on gas).

**Modeling choice.** The number of boxes  $n_g$  determines precision:

- $n_g = 1$ : a single decay process (e.g.  $\text{CH}_4$ ,  $\text{N}_2\text{O}$ ).
- $n_g = 2-3$ : separates atmosphere, fast reservoirs (biosphere/upper ocean), and slow reservoirs (deep ocean) for  $\text{CO}_2$ .
- Larger  $n_g$ : more realistic sequestration processes (soil carbon, permafrost, multiple ocean layers) at the cost of extra parameters.

### Conservation:

- the system is physical, so total mass is conserved: transfers occur *between* boxes, with no loss outside the system (except for explicitly modelled sinks).

### CO<sub>2</sub> : three-box carbon cycle

$$\mathbf{M}_{t+\Delta}^{CO_2} = \Phi^{CO_2} \mathbf{M}_t^{CO_2} + \Delta \Xi^{CO_2} E_t^{CO_2}, \quad \mathbf{M}_t^{CO_2} = \begin{bmatrix} M_{AT} \\ M_{UP} \\ M_{LO} \end{bmatrix} \in \mathbb{R}^3$$

- $\Phi^{CO_2} \in \mathbb{R}^{3 \times 3}$ : transfer matrix. Diagonal: persistence ( $< 1$ ). Off-diagonal: transfers across boxes.
- $\Xi^{CO_2} \in \mathbb{R}^{3 \times 1}$ : injection vector (typically only atmosphere receives emissions).
- Boxes: **AT** = atmosphere, **UP** = upper ocean/biosphere, **LO** = deep ocean.



$\text{CH}_4$  : one-box decay to baseline

$$M_{t+\Delta}^{\text{CH}_4} = M_t^{\text{CH}_4} - \Delta \delta_{\text{CH}_4} (M_t^{\text{CH}_4} - M_0^{\text{CH}_4}) + \Delta \kappa_{\text{CH}_4} E_t^{\text{CH}_4}$$

- $M_0^{\text{CH}_4}$ : preindustrial baseline ( $\approx 722$  ppb).
- $\delta_{\text{CH}_4}$ : decay rate ( $\tau \sim 12$  y lifetime).
- $\kappa_{\text{CH}_4}$ : conversion emissions  $\rightarrow$  ppb/yr ( $1 \text{ ppb} \approx 2.78 \text{ MtCH}_4$ ).

# Radiative Forcing

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## Forcing: Definition

[Emissions (flows)]  $\rightarrow$  [Concentration (stock)]  $\rightarrow$  [Forcing (flow)]  $\rightarrow \dots$

**Key idea.** Radiative forcing is the *side effect of rising concentrations*: it increases the thickness of the atmospheric “blanket” that traps heat.

- **Baseline:** in the preindustrial climate, incoming solar  $\approx$  outgoing infrared (energy balance).
- **Forcing:** any perturbation that shifts this balance.
  - Positive forcing  $\Rightarrow$  warming (extra blanket from GHGs).
  - Negative forcing  $\Rightarrow$  cooling (reflective aerosols, volcanic dust).
- **Units.** Watts per square meter ( $\text{W}/\text{m}^2$ ).

# Forcing: General Formula

**Key idea.** Total forcing is the sum of contributions from different agents, each measured relative to a natural (preindustrial) baseline.

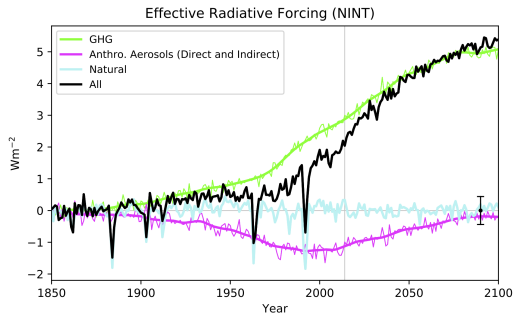
## General formulation

$$F_t = \sum_{g \in \{\text{CO}_2, \text{CH}_4, \text{N}_2\text{O}, \text{others}\}} F_t^g$$

$$F_t^g = f_g(M_t^g) - f_g(M_0^g)$$

- $M_t^g$  = atmospheric concentration of gas  $g$  at time  $t$ .
- $M_0^g$  = preindustrial reference level (e.g. 278 ppm CO<sub>2</sub>, 722 ppb CH<sub>4</sub>).
- $f_g(\cdot)$  = functional form mapping concentration to forcing.
  - CO<sub>2</sub>: logarithmic in concentration.
  - CH<sub>4</sub>, N<sub>2</sub>O: square-root formulations (with overlap correction).
  - Other agents: aerosols (negative forcing), solar variability, land-use.

## Forcing: Some illustration



- **Pink line:** Anthropogenic aerosols  $\Rightarrow$  *negative forcing*.  
e.g. sulfate aerosols (from burning coal and oil-  $\text{SO}_2$  emissions), very reflective: scatter sunlight back to space, short lifetime (days-weeks).
- **Cyan line:** Natural forcings (solar variability, volcanic eruptions).

## Forcing: CO<sub>2</sub> Forcing

Forcing formulas map concentrations into energy imbalance (W/m<sup>2</sup>), measured relative to preindustrial levels.

### CO<sub>2</sub> forcing

$$F_t^{CO_2} = F_{2 \times CO_2} \cdot \frac{\ln(M_t^{CO_2}/M_0^{CO_2})}{\ln 2}$$

- Preindustrial  $M_0^{CO_2} = 278$  ppm
- Forcing at doubling:  $F_{2 \times CO_2} \approx 3.7$  W/m<sup>2</sup>
- Today ( $M_t^{CO_2} \approx 420$  ppm):

$$F_t^{CO_2} \approx 3.7 \cdot \frac{\ln(420/278)}{\ln 2} \approx 2.1 \text{ W/m}^2$$

### CH<sub>4</sub> forcing

$$F_t^{CH_4} = \alpha_{CH_4} \left( \sqrt{M_t^{CH_4}} - \sqrt{M_0^{CH_4}} \right)$$

- Preindustrial  $M_0^{CH_4} = 722$  ppb
- Forcing coefficient  $\alpha_{CH_4} \approx 0.036 \text{ W/m}^2/\sqrt{\text{ppb}}$
- Today ( $M_t^{CH_4} \approx 1860$  ppb):

$$F_t^{CH_4} \approx 0.036 \cdot (43.1 - 26.9) \approx 0.6 \text{ W/m}^2$$

## Temperatures (output)

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[Emissions (flows)]  $\rightarrow$  [Concentration (stock)]  $\rightarrow$  [Forcing (flow)]  $\rightarrow$  [Temperature (stock)]

- **Key point:** Radiative forcing  $\neq$  instantaneous warming.

Temperatures behave as a *state variable*: they accumulate the effect of past forcings, with strong inertia.

- **Sources of inertia:**
  - The ocean absorbs most of the excess heat, releasing it slowly.
  - Ice sheets and cryosphere respond on multi-decadal to millennial scales.
  - Feedbacks (clouds, carbon cycle) operate with delays.
- **Illustration:** Even if CO<sub>2</sub> concentrations stabilized today, global temperatures would still rise for decades (“committed warming”).

## Temperatures: Warming as an ODE

As with concentrations, temperatures evolve as a linear dynamic system: forcing is a flow, temperatures are stocks.

### General formulation

$$\mathbf{T}_t = \mathbf{T}_t + \Delta \left( \mathbf{A}_T \mathbf{T}_t + \mathbf{B}_T F_t \right), \quad \mathbf{T}_0 = \mathbf{0}$$

- $\mathbf{T}_t$  = vector of temperature anomalies (e.g. atmosphere, upper ocean, deep ocean).
- $F_t$  = total radiative forcing at time  $t$  (flow).
- $\mathbf{A}_T$  = matrix of heat exchange and feedback parameters.
- $\mathbf{B}_T$  = vector linking forcing to temperature.
- $\mathbf{T}_0 = \mathbf{0}$ : preindustrial baseline (no initial warming).

## Temperatures: a two box model

A simple reduced-form system tracks two temperature stocks: atmosphere/surface and deep ocean:

$$T_{t+\Delta}^{AT} = T_t^{AT} + \Delta \left[ c_1 F_t - c_1 \frac{F_{2 \times CO_2}}{T_{2 \times CO_2}} T_t^{AT} - c_1 c_3 (T_t^{AT} - T_t^{LO}) \right]$$

$$T_{t+\Delta}^{LO} = T_t^{LO} + \Delta \left[ c_4 (T_t^{AT} - T_t^{LO}) \right]$$

- $T^{AT}$ : atmospheric/surface temperature anomaly (fast response).
- $T^{LO}$ : deep-ocean temperature anomaly (slow response, high inertia).
- $F_t$ : total radiative forcing at time  $t$ .
- Parameters:  $c_1$  (climate response rate),  $c_3$  (surface–ocean coupling),  $c_4$  (deep-ocean heat uptake),  $T_{2 \times CO_2}$  (climate sensitivity).
- Initial condition:  $T_0^{AT} = T_0^{LO} = 0$  (preindustrial baseline).

# Temperatures: asymptotic dynamics

Model in *relative terms* with respect to a doubling  $\text{CO}_2$  and hold it fixed (no other forcings).

- **Forcing:**

$$M_{AT}^{CO_2} = 2M_{AT,0}^{CO_2} \Rightarrow F^{CO_2} = F_{2\times CO_2} \approx 3.6 \text{ W/m}^2$$

- **Dynamics (discrete-time 2-box model):**

$$T_{t+\Delta}^{AT} = T_t^{AT} + \Delta \left[ c_1 F_{2\times CO_2} - c_1 \frac{F_{2\times CO_2}}{T_{2\times CO_2}} T_t^{AT} - c_1 c_3 (T_t^{AT} - T_t^{LO}) \right]$$

$$T_{t+\Delta}^{LO} = T_t^{LO} + \Delta c_4 (T_t^{AT} - T_t^{LO})$$

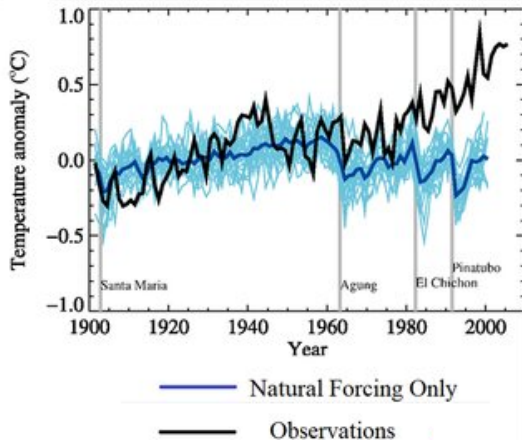
- **Equilibrium:**

$$T^{AT} = T^{LO} = T_{2\times CO_2} \approx 3.1^\circ\text{C}$$

**Takeaway:**  $F_{2\times CO_2}$  = forcing from doubling,  $T_{2\times CO_2}$  = long-run warming response.

They anchor reduced-form climate dynamics.

# Temperatures: Human vs. Natural Forcings



## To wrap up

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To summarize the result, consider a carbon pulse of 1GtC:

$+1GtC \rightarrow +0.47 \text{ ppm } CO_2 \rightarrow +0.01 \text{ W/m}^2 \text{ forcing} \rightarrow +0.006^\circ C \text{ in the long term}$

Thank you!

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